Midterm Exam

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Question 1

As we are testing for a precise null in a continuous distribution, we need to use a point mass. Setting a hypothesis of $H_0 : \theta = 0, H_1 : \theta \neq 0$.

$$p(\theta) = 0.5\phi(\theta; 0, 1) + 0.5\delta_0(\theta) = p_1\phi(\theta; 0, 1) + p_0\delta_0(\theta)$$

where $p_0 = p_1 = 0.5$.
To determine the posterior distribution, first the marginal has to be found, which is provided as:

$$m(y) = \int_{-\infty}^{\infty} p(y|\theta)p(\theta)d\theta$$

where the normalising constant for y=0 is given as:

$$m(0) = 0.5\int_{-\infty}^{\infty} \phi^2(\theta; 0, 1)d\theta + 0.5\phi(0; 0, 1)$$

which we can conceptualise as:

$$m(0) = 0.5m_1 + 0.5m_0$$

where m_0, m_1 are the marginals associated with H_0, H_1 respectively.
The posterior distribution probability at θ_0 is defined as:

$$p(\theta_0|y) = \frac{p(y|\theta_0)p_0}{m(y)}$$

At y=0:

$$p(\theta_0|y = 0) = \frac{p(y|\theta_0)(0.5)}{m(0)}$$

Evaluating m(0) first using the PDF of a normal distribution: $f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$m(0) = 0.5(m_1) + 0.5(m_0) = \frac{0.5}{2\pi}\int_{-\infty}^{\infty} e^{-\theta^2}d\theta + \frac{0.5}{\sqrt{2\pi}}$$

Refer to the code file for the definite integral calculation of $\int_{-\infty}^{\infty} e^{-\theta^2}d\theta$ using the SymPy module in Python, which finds this integral equates to $\sqrt{\pi}$:

$$m(0) = 0.5(\frac{\sqrt{\pi}}{2\pi}) + 0.5(\frac{1}{\sqrt{2\pi}})$$

where $m_1 = \frac{\sqrt{\pi}}{2\pi}$ which we will use later.

$$m(0) = \frac{\sqrt{\pi}}{4\pi} + \frac{1}{2\sqrt{2\pi}} = \frac{1}{2\sqrt{\pi}}(\frac{1}{2} + \frac{1}{\sqrt{2}}) \approx 0.3405$$

Now evaluating the likelihood at θ_0:

$$p(y = 0|\theta_0) = \frac{1}{\sqrt{2\pi}} e^0 = \frac{1}{\sqrt{2\pi}} \approx 0.3989$$

Back to the posterior distribution at $\theta_0, y = 0$:

$$p(\theta_0|y) = \frac{0.3989(0.5)}{0.3405} \approx 0.5858$$

Now that we have the posterior probability (weight) at the point mass, we need to describe the rest of the continuous distribution.
The posterior weight for the rest of the distribution:

$$p(\theta_{\theta \neq 0}|0) = 1 - p(\theta_0|0) \approx 0.4142$$

We can also calculate the Bayes Factor here using our marginal from above:

$$B_{01} = \frac{p(y|\theta_0)}{m_1} = \frac{\frac{1}{\sqrt{2\pi}}}{\frac{\sqrt{\pi}}{2\pi}} = \frac{2}{\sqrt{2}} = 1.414$$

According to conjugacy rules, as the prior and likelihood are both normally distributed, the posterior will also follow a normal distribution. According to the conjugacy table, the posterior mean will be the same as the likelihood mean of 0:

$$\mu_{post} = \frac{\tau^2 + \sigma^2 \mu}{\tau^2 + \sigma^2} = 0$$

In a similar fashion, we can again refer to the conjugacy table to find the posterior variance update. Given that are component distributions all have variance of 1:

$$\tau^2_{post} = \frac{\sigma^2 \tau^2}{\sigma^2 + \tau^2} = \frac{1}{2}$$

Hence:

$$\theta_{\theta \neq 0}|y \sim N(0, \frac{1}{2})$$

Therefore our mixture posterior distribution is:

$$p(\theta|y) = (0.5858)\delta_0(\theta) + (0.4142)p(\theta_{\theta\neq 0}|y; 0, \frac{1}{2})$$

So in our posterior distribution the weights have changed so that the probability that $\theta = 0$ has increased. This makes intuitive sense as our observed measurement was $y = 0$, so we expect this probability to rise.

Question 2

2.1)

Before any data is observed, the prior predictive distribution uses only prior knowledge to make a prediction. It is also known as the marginal and can be defined as:

$$P(Y) = \int P(Y|\theta)\pi(\theta)d\theta$$

In our case $\pi(\theta)$ is a uniform distribution that doesn't change the prior predictive. The probability density function of the likelihood given it's a geometric function is:

$$P(Y = y|\theta) = \theta(1-\theta)^{y-1}$$

$$P(Y \geq y|\theta) = (1-\theta)^{y-1}$$

As we want to find the prior probability of smoke-free at least 7 days, we can construct the prior predictive distribution with limits [0,1] as θ is a probability:

$$P(Y \geq y = 7) = \int_0^1 P(Y \geq y = 7|\theta)d\theta$$

$$P(Y \geq y = 7) = \int_0^1 (1-\theta)^{7-1}d\theta = \int_0^1 (1-\theta)^6 d\theta$$

This can be solved algebraically by substitution where $u = 1 - \theta$ and $d\theta = -du$:

$$\int_0^1 (1-\theta)^6 d\theta = \int_0^1 -(u)^6 du = \left[\frac{u^7}{7}\right]_0^1 = \frac{1}{7}$$

Therefore the probability the smoker will be smoke-free for at least 7 days is $\frac{1}{7}$ or ≈ 0.143. For reference, this integral has also been run in the code file using SymPy.

2.2)

Now given that we have an observed measurement of $Y = 5$ we can construct a posterior predictive distribution of Y. First, to create the posterior distribution we can use:

$$\pi(\theta|Y) = \frac{P(Y|\theta).\pi(\theta)}{P(Y)}$$

The denominator is the marginal, which is equivalent to the prior predictive distribution defined above, so plugging in the known functions:

$$\pi(\theta|Y=y) = \frac{\theta(1-\theta)^{y-1}}{\int_0^1 \theta(1-\theta)^{y-1}d\theta}$$

Integral can be solved via integration by parts, setting $u=\theta, du=d\theta$ and $dv=(1-\theta)^4 d\theta, v=-\frac{(1-\theta)^5}{5}$

$$\int_0^1 \theta(1-\theta)^4 d\theta = \left[-\frac{\theta(1-\theta)^5}{5}\right]_0^1 - \int_0^1 -\frac{(1-\theta)^5}{5}d\theta = 0 - \left[\frac{(1-\theta)^5}{5*6}\right]_0^1 = \frac{1}{30}$$

$$\pi(\theta|Y=5) = 30\theta(1-\theta)^4$$

Posterior distribution resembles a Beta distribution. Using the kernel method we can find it's parameters:

$$\pi(\theta|Y=5) \propto \theta(1-\theta)^4 \propto \theta^{a-1}(1-\theta)^{b-1}$$

$$a=2, b=5$$

$$\pi(\theta|Y=5) \sim Beta(2,5)$$

2.3)

The posterior predictive distribution is distribution of a future observation y_*, given the observations we have already seen:

$$P(y_*|Y=5) = \int_0^1 P(y_*|\theta)\pi(\theta|Y=5)d\theta$$

$$P(y_*|Y=5) = \int_0^1 (1-\theta)^{y_*-1}.(30)\theta(1-\theta)^4 d\theta = 30\int_0^1 \theta(1-\theta)^{y_*+3}d\theta$$

The closed form for this distribution has not been calculated using the Beta function identity (although it is possible to solve), as it is not needed for part 4 below. The integral above fully defines the distribution and is a simpler solution when ingesting new observations.

2.4)

To find the probability of at least 7 smoke free days then:

$$P(y_* \geq 7|Y=y) = \int_0^1 P(y_* \geq 7|\theta)\pi(\theta|Y=5)d\theta$$

$$P(y_* \geq 7|Y=5) = 30\int_0^1 (1-\theta)^{7-1}.\theta(1-\theta)^4 d\theta = 30\int_0^1 \theta(1-\theta)^{10}d\theta$$

The integral can once again be solved via integration by parts, setting $u = \theta, du = d\theta$ and $dv = (1-\theta)^{10} d\theta, v = -\frac{(1-\theta)^{11}}{11}$

$$\int_0^1 \theta(1-\theta)^{10} d\theta = \left[-\frac{\theta(1-\theta)^{11}}{11}\right]_0^1 - \int_0^1 -\frac{(1-\theta)^{11}}{11} d\theta = 0 - \left[\frac{(1-\theta)^{12}}{11*12}\right]_0^1 = \frac{1}{132}$$

Please refer to the code file for the alternative method to calculate these integrals, solved using the SymPy package in Python.
Therefore, given the first attempt was 5 days, the probability of the second attempt being at least 7 days is:

$$P(y_* \geq 7|Y=5) = \frac{30}{132} = 0.227$$

Question 3

3.1)

In order to find the posterior distribution, all the conditional distributions are required for Gibbs sampling. To find the conditional distribution, first we need to compute the joint distribution, which would be the likelihood multiplied by the prior for both σ^2 and b.

In order to avoid confusion with the squared term in sigma, we will be defining $x = \sigma^2$ from now on. The vector $\bar{y}, \bar{x}$ will denote all possible values of y, x respectively.

Joint distribution:

$$p(\bar{x}, b|\bar{y}) = p(\bar{y}|\bar{x}).\,p(\bar{x}|b).\,p(b)$$

Prior for b can be constructed from the PDF of an exponential distribution:

$$p(b|\lambda = 1) = \lambda e^{-\lambda b} = e^{-b}$$

Prior for $x = \sigma^2$ constructed using the PDF of an Inverse-Gamma distribution:

$$p(\bar{x}|a = 10, b) = \frac{b^{10}}{\Gamma(10)} \bar{x}^{-11} e^{-b/\bar{x}} = \prod_{i=1}^{n} \frac{b^{10}}{\Gamma(10)} x_i^{-11} e^{-b/x_i}$$

Likelihood term is a normal distribution described by:

$$p(\bar{y}|\mu = 0, \bar{x}) = \sqrt{\frac{1}{2\pi\bar{x}}} e^{-\bar{y}^2/2\bar{x}}$$

Combining these distributions back into the joint distribution:

$$p(\bar{x}, b|\bar{y}) = (\sqrt{\frac{1}{2\pi\bar{x}}} e^{-\bar{y}^2/2\bar{x}}).\,\frac{b^{10}}{\Gamma(10)} \bar{x}^{-11} e^{-b/\bar{x}} = \prod_{i=1}^{n} \frac{b^{10}}{\Gamma(10)} x_i^{-11} e^{-b/x_i}.\,e^{-b}$$

$$p(\bar{x}, b|\bar{y}) = \frac{1}{\Gamma(10)\sqrt{2\pi}} . \bar{x}^{-11-0.5} \prod_{i=1}^{n} b^{10} e^{-b/x_i} e^{-b}$$

Conditional of b

Isolate terms containing b and expand the $\bar{x}$ vector into it's components. We want to expand all the components as b is a single value so we need to combine all the x terms:

$$p(b|\bar{x}, \bar{y}) = e^{-b} \prod_{i=1}^{n} b^{10} e^{-bx_i} = e^{-b} b^{10n} e^{-b \sum_i \frac{1}{x_i}} = b^{10n} e^{-b(1+\sum_i \frac{1}{x_i})}$$

As this resembles a Gamma distribution, we can compare indices using the kernel method for the known Gamma PDF:

$$b^{\alpha-1} e^{-\beta b} \propto b^{10n} e^{-b(1+\sum_i \frac{1}{x_i})}$$

$$\alpha = 10n + 1, \quad \beta = \sum_i \frac{1}{x_i} + 1 = \sum_i \frac{1}{\sigma_i^2} + 1$$

Therefore the conditional distribution of b is:

$$b|\bar{\sigma^2}, \bar{y} \sim Ga(10n + 1, \sum_i \frac{1}{\sigma_i^2} + 1)$$

Conditional of $\bar{x} = \bar{\sigma}^2$

$$p(\bar{x}|b, \bar{y}) = \bar{x}^{-11.5} e^{-\frac{\bar{y}^2}{2\bar{x}} - b/\bar{x}} = \bar{x}^{-11.5} e^{-(\bar{y}^2/2+b)\frac{1}{\bar{x}}}$$

This resembles an Inverse Gamma distribution and we can use the kernel method:

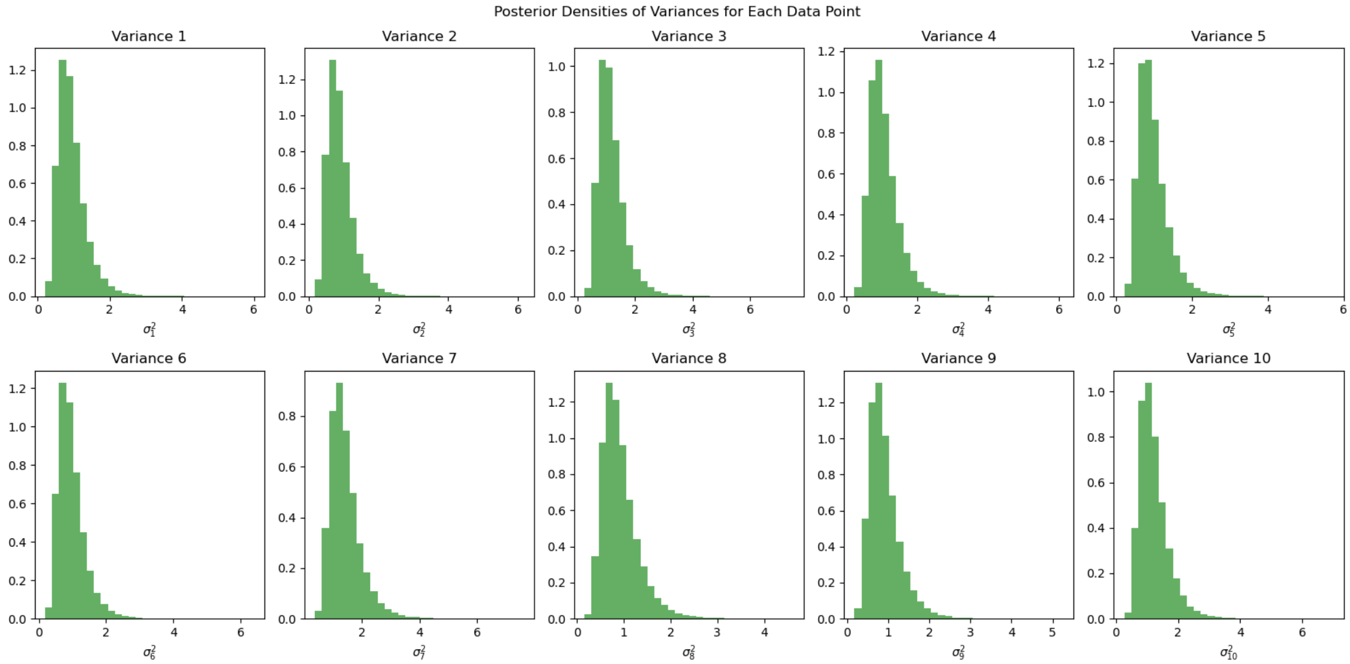
$$x^{-a-1} e^{-\frac{b}{x}} \propto \bar{x}^{-11.5} e^{-(\bar{y}^2/2+b)\frac{1}{\bar{x}}}$$

$$a_{\sigma^2} = 10.5, \quad b_{\sigma^2} = \frac{\bar{y}^2}{2} + b$$

Therefore the conditional distribution of $\bar{\sigma}^2$ is:

$$\bar{\sigma}^2|b, \bar{y} \sim Inv - Gamma(10.5, \frac{\bar{y}^2}{2} + b)$$

Please refer to the code file for implementation of Gibbs sampling.



3.2)

Parameter	Posterior Mean
σ_1	0.953
σ_2	0.909
σ_3	1.192
σ_4	1.063
σ_5	0.963
σ_6	0.973
σ_7	1.417
σ_8	0.919
σ_9	0.912
σ_{10}	1.211

3.3)

Posterior mean for b = 8.644

Calculated using the ArviZ Python package, the 95% HPD credible interval for b = [4.968, 12.738]

Other Distribution Definitions Used

Gamma Distribution

- **Notation:** $X \sim Ga(\alpha, \beta)$
- **Parameters:** α (shape), β (rate)
- **PDF:** $f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$

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Name Nedeesha Weerasuriya

Problem	1	2	3	Total
Score	/35	/35	/30	/100